

THE INTERN DIARIES

Intern Report

Product Basket Analysis

WEEK 4 | DAILY JOURNAL ENTRY



LEARNING BY DOING, ONE DAY AT A TIME

Online Retail II Dataset (MARKET BASKET ANALYSIS)

1. Which products should be recommended together?

Answer: Recommend products that have **high Confidence** and **high Lift**. These are items where purchasing Product A makes purchasing Product B significantly more likely than random chance. Focus on rules where the "If_Buys" product is a common entry point (high Support) and the "Then_Also_Buys" product has a strong conditional probability. From the **association_rules.csv** file we can see that there are nearly **244** products in the retail store that have high confidence as well as lift which can be recommended together.

2. Which products could be bundled?

Answer: Bundle products that have **high Support AND high Lift**. High Support means the pair appears frequently enough across many orders to justify creating a physical or digital bundle (e.g., "Party Starter Kit"). High Lift ensures the bundle isn't just two random popular items, but a genuine complementary pair. Avoid bundling items with low Support, as inventory will sit unsold.

With a **support** value between **(1-5)%** and **lift > 3** this remain the top products that can be bundled together.

3. Which combinations are suitable for cross-selling?

Answer: Cross-selling is about suggesting an *additional* item at checkout ("Would you like to add X?"). You want pairs where buying Product A strongly predicts buying Product B, but B is not a "must-have" companion (like batteries for a toy). Look for rules with **high Confidence** but **moderate Support**. High confidence ensures the recommendation is relevant; moderate support ensures it's not too obvious/common (which wastes screen real estate).

- **Confidence > 0.4 (40%) AND Support between 0.01 and 0.05 (1%–5%).**
- **Sort by Lift descending.**

These items are frequently bought together enough to trigger a recommendation, but not so ubiquitous that everyone already knows about them.

4. Which products may benefit from joint promotions?

Answer: Joint promotions (e.g., "Buy A + B, get 10% off") work best when the association is strong (**High Lift**) but the current co-purchase rate isn't yet maximal (**Moderate Confidence**). If confidence is already 90%, they don't need a discount—they're already bundled mentally. You want to push customers over the edge who buy A but hesitate on B.

- **Lift > 3** (strong relationship).
- **Confidence < 0.6 (60%)** (room for improvement).
- **Support descending** (potential pairs)

5. Which products should be displayed near each other?

Answer: Physical or digital shelf placement relies on **high-frequency co-occurrence**. You want items that appear together in many baskets (**High Support**) regardless of individual popularity. This optimizes picking efficiency (warehouse) or impulse buys (retail). Ignore lift here—focus purely on volume of co-purchases.

- **Support descending**.
- Take the **Top 10–15 pairs**.

These are the "power couples" of your catalog. Placing them adjacently reduces search time and captures natural shopping flows.

6. Which product relationships are strong enough to investigate further?

Answer: This is for anomaly detection and strategic insights. Look for extremely high Lift (> 5 or > 10) even if Support is low. These rare but powerful associations often reveal niche use cases, hidden bundles, or data quirks worth exploring. They might indicate a new market segment (e.g., "wedding favors" cluster) or a data error (e.g., duplicate SKUs).

- **Lift descending**.
- Filter for **Lift > 5**.
- **Support** column for these top rows—if it's very low (< 0.01)

PRASANNA SHETTY, 06-OCT-2026

High lift = surprising strength. Low support = rarity. Together, they signal "something unusual is happening here." Is it a new trend? A bulk-buy pattern? A coding error?